

Design and Development of an Integrated Management System Using Linux Fedora

Group Members:

Nida Munawar Ali (68453)

Rija Munawar Ali (68452)

Course Name:

Operating System Lab

Submitted to:

Ali Hassan Sial

Department of:

Software Engineering

Iqra University(FEST)

1. Background/Theory:

An operating system is responsible for managing the hardware and software resources of a computer and providing a suitable environment for users and applications. Linux Fedora is an open-source operating system that provides various built-in utilities for managing processes, files, system resources, scheduled tasks, and users.

The proposed project is an Integrated Management System using Linux Fedora that will combine five important management functions into one system. These functions are System Resource Management, Process Management, File Management, Task Automation, and User Management. The purpose of integrating these functions is to demonstrate how different operating system responsibilities can be managed through a single organized system.

The System Resource Manager will monitor CPU usage, memory usage, and disk space to provide information about the current utilization of system resources. The Process Management System will allow users to view, filter, monitor, and manage running processes. The File Management System will provide operations for managing files and directories, including searching, organizing, creating backups, and controlling file permissions.

The Task Automation Manager will be used to schedule repetitive tasks according to user-defined requirements, such as performing regular backups. The User Management System will handle user accounts and permissions, helping control access to system resources. Together, these modules will demonstrate important operating system concepts such as resource management, process control, file systems, scheduling, and security.

2. Procedure / Methodology:

The project will be developed through a systematic process. First, the requirements and functionalities of the five management modules will be identified. After identifying the requirements, the overall structure of the Integrated Management System will be designed so that all five modules can operate independently while remaining part of the same system.

The next step will be the implementation of each module using Linux Fedora utilities and shell scripting. The System Resource Manager will be developed for resource monitoring, the Process Management module for process monitoring and control, the File Management module for file and directory operations, the Task Automation module for scheduling tasks, and the User Management module for user accounts and permissions.

After implementation, each module will be tested individually to identify and correct errors. Finally, all five modules will be integrated and tested as a complete system. The final evaluation will determine whether the system performs the required management operations correctly and efficiently.

